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dsquasar

Backing Up the GDEX Archive onto Quasar

Gather → List → Tar → Transfer, tracked end-to-end in GDEXDB

Backup & Disaster Recovery

Batch Operations

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Tars through `dsarch AQ` • uploads through `dsglobus` • scheduled by `dscheck`

What is dsquasar?

dsquasar keeps a **tape copy of the whole GDEX archive**. It finds files never backed up, bundles them into large tar files with **dsarch AQ**, then uploads many tars at once to the Quasar endpoints with **dsglobus**.

Driven entirely by GDEXDB flags

- Nothing is configured per run: a dataset joins the backup when `dataset.backflag` is set; `dsgroup.backflag` refines it per group.
- Any file with `bid = 0` is not backed up yet, and is picked up automatically.
- Every tar gets a `bfile` record whose status tracks it from list to tape.
- Work is resumable: a run that stops leaves its pending records behind.



Gather

find Web/Saved files still to back up



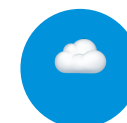
List

write dsarch input files, add bfile records



Tar

bundle into ≥ 5 GB tar files via dsarch



Transfer

dsglobus batches to gdex-quasar (+drdata)



Verify

compare sizes on tape, reset if wrong



Report

statistics & email for specialists

Install & Resources



GitHub Repository

Source code, issues, and contributions.

```
git clone https://github.com/  
NCAR/rda-python-dsquasar.git
```

→ github.com/NCAR/rda-python-dsquasar



PyPI Package

Install the released package with pip; pulls in rda_python_common and rda_python_dsarch.

```
pip install rda_python_dsquasar
```

→ pypi.org/project/rda-python-dsquasar



Built-in Help

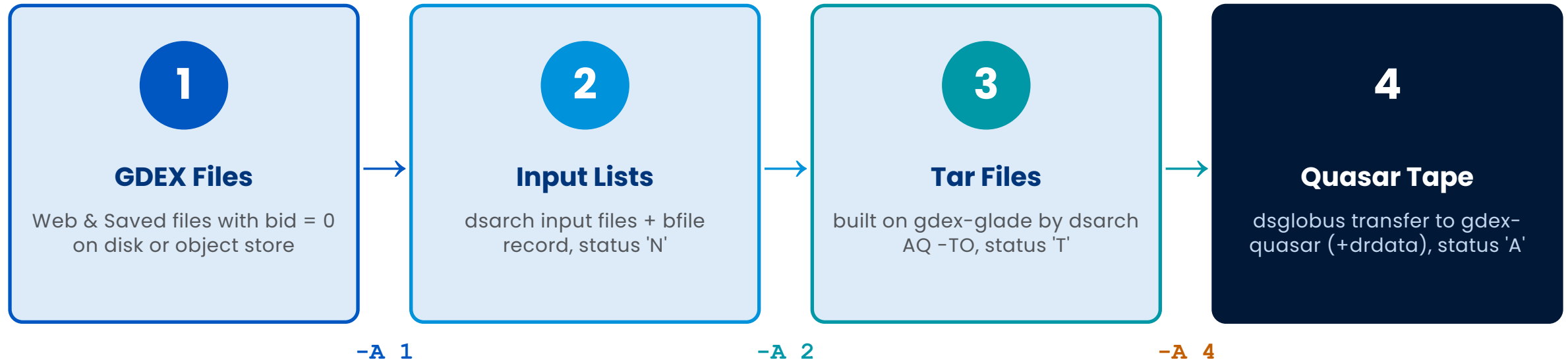
The full usage document ships with the package and is shown by -h.

```
dsquasar -h  
# renders dsquasar.usg
```

→ src/rda_python_dsquasar/dsquasar.usg

one console command: `dsquasar` backs up the whole GDEX archive onto the Quasar end points.

The Four-Stage Backup Pipeline



-A 3 is the workhorse. It runs stages 1 and 2 together: files listed this run are tarred immediately, and existing 'N' records left by an earlier run are picked up too. **-A 4** then moves the tars to tape. Use **-A 7** for the full chain in one run.

What dsquasar Reads and Writes in GDEXDB

Backup Flags

dataset / dsgroup **read**

- backflag 'B' — Backup copy only
- backflag 'D' — Backup + Drdata copy
- backflag 'N' — do not back up
- group flag 'P' inherits the dataset flag
- dataset.pid / lockhost — the run lock

Data Files

wfile / sfile **read + update**

- bid = 0 ⇒ not backed up yet
- bid = <bfile> ⇒ inside that tar
- data_size / checksum / timestamps re-verified on disk before tarring
- locflag 'O' files are pulled from object storage first

Backup File Record

bfile **write**

- one record per tar file
- status 'N' → 'T' → 'A'
- type 'B' or 'D'; dsids lists extra datasets
- note holds the whole input-file listing — the manifest of what is inside

The bfile note is the recovery key: it records every member file, its size and checksum, so a tar can be re-listed, re-tarred, or restored without re-reading the archive.

Anatomy of a Command

```
dsquasar (-a | -t dNNNNNN...) [-A Bits] [-B|-D] [Run...] [-e|-E]
```

Scope

required

Which datasets. -a walks every dataset with a Backup flag; -t takes IDs and accepts the SQL wildcard %.

Action

-A Bits

What to do, as a bitmask: 1 list, 2 tar, 4 transfer, 8 check, 16 stats. Defaults to 3.

Copies

-B | -D

Which end points. -B Backup only, -D Backup + Drdata. Default: both.

Run control

-b -d -m -l -n -W -w -u

How it runs: background, PBS delay, processes, locking, dry run, worker slots.

Notes either -a or -t must be given, otherwise the usage is shown • -a wins over -t with a warning • the run always chdir's to the shared Quasar work directory • -h prints the full usage document.

Six Commands That Cover Most Work

1 See what is waiting

Counts only — gathers and reports, performs no backup.

```
dsquasar -t d123456 -n
```

2 Back up one dataset

Create the input lists and build the tar files.

```
dsquasar -t d123456 -A 3
```

3 Send the tars to tape

Batches built tars into ≥ 90 GB dsglobus transfers.

```
dsquasar -t d123456 -A 4
```

4 The nightly production run

All flagged datasets, as a PBS batch job with email.

```
dsquasar -a -A 3 -e -b -d PBS
```

5 Report on everything

Backup statistics for all datasets, mailed to the specialist.

```
dsquasar -a -A 16 -e
```

6 Clear a stale lock

Unlock datasets left locked by a crashed run.

```
dsquasar -t d123456 -u
```

Everything is incremental. A file is a candidate only while `bid = 0`, so re-running never redoes finished work.

Backup Flags: What Gets Two Copies

A dataset is put under Quasar backup by setting `dataset.backflag` — that one field is the switch, nothing else enrolls a dataset.

Flag	Value in <code>dataset.backflag</code> / <code>dsgroup.backflag</code>
'B'	Quasar Backup only — endpoint <code>gdex-quasar</code>
'D'	Backup + Drdata — also <code>gdex-quasar-drdata</code>
'N'	Not backed up at all
'P'	Group level only: inherit the dataset's flag

Group flags refine, they do not enroll. Once the dataset flag is set, a top-level `dsgroup` flag is applied group by group.

Set Backup Flag in the web config tool gdex.ucar.edu/rda_pg_config/

DSARCH → Dataset Information

sets `dataset.backflag` for the whole dataset

DSARCH → Group Information

sets `dsgroup.backflag` for one group

Per-run filter:

–B or –D limits a run to what is flagged that way; with neither, each dataset follows its own flag.

Two Globus end points, one source

gdex-glade
tar files built on disk



gdex-quasar

Backup copy — always written

gdex-quasar-drdata

Disaster-recovery copy — only for flag 'D'

A 'D' backup writes Drdata first, then Backup; both must succeed before the local tar is deleted.

The -A Action Bitmask

-A 1**Create Input**

list Web/Saved files, add bfile records 'N'

-A 2**Tar Backup**

build ≥ 5 GB tars from 'N' lists $\rightarrow$ 'T'

-A 3**Create + Tar**

the default, and the normal production run

-A 4**Transfer**

dsglobus batches of 'T' tars to Quasar $\rightarrow$ 'A'

-A 6**Tar + Transfer**

build then ship in one run

-A 7**Full chain**

list, tar, and ship end to end

-A 8**Check**

compare sizes on tape, reset bad ones to 'N'

-A 16**Statistics**

dump counts for done and pending work

It really is a bitmask. $3 = 1+2$, $6 = 2+4$, $7 = 1+2+4$ — only those combinations are accepted. 1, 8 and 16 are single-process bookkeeping actions; 2, 3, 4, 6, 7 are the batch actions that get PBS cpus and multi-processing.

The bfile Status Lifecycle



Records also move backwards — that is the self-healing part

'T' → 'N'

the local tar file is missing when a transfer is prepared, so the tar is rebuilt from the stored input lists

'A' → 'N'

`-A 8` finds the file absent or the wrong size on the end point, so the whole tar is redone

record dropped

dsarch reports every member file as already backed up elsewhere, so the duplicate placeholder is deleted

Because status drives everything, an interrupted run is never a problem: whatever is still **'N'** or **'T'** is simply picked up by the next run.

-A 1 Deep Dive: Building the Input Lists

For each flagged dataset

1. Lock the dataset, then re-gather its files under the lock
2. Skip files already listed in an open 'N' record
3. Re-check each file on disk or object storage; fix size, checksum and timestamp in GDEXDB
4. Append a line per file to a dsarch input file, per type
5. Close the group when big enough; add the bfile record

Input file layout

```
DS<=>d123456
WF<:>WT<:>SZ<:>MC<:>
file.nc<:>data<:>2147483648<:>3f1c...<:>
```

When is a tar big enough?

Threshold	Rule applied while accumulating files
20 GB	a single file alone is enough to be its own tar
5 GB	normal target size while under 100 files
2 GB	accepted minimum once 100 files have piled up
65 datasets	hard cap on datasets sharing one tar file

Small files are the reason. Tape wants few large files, so many tiny files are packed together — across datasets if need be — while a single very large file is never split.

Resulting tar file name

```
G001/d123456_sn40321_dn3_fn218.tar
```

subpath / dsid _ bid _ dataset count _ file count

-A 2 Deep Dive: Building the Tar Files

```
dsarch d123456 AQ -QT D -QF G001/d123456_sn40321_dn3_fn218.tar -TO -OE -MD -IF d123456_W_40321.txt ...
```

AQ Archive Quasar — pack the listed files into one tar

-QT copy type: 'B' Backup only, or 'D' Backup + Drdata

-QF the tar file name recorded in the bfile record

-TO Tar Only — build the tar, do not upload it

-OE overwrite an existing Quasar file of that name

-MD act on any specialist's dataset

-IF the input files listing what goes into the tar

One child per tar file

With -m N, each tar file is handed to its own child process, so tarring scales widely across cpus.

Missing lists are rebuilt

The bfile note holds a copy of every input file, so a list deleted from the work directory is written back out before tarring.

Verified before it counts

After the children finish, the confirmed counts are re-read from GDEXDB, so the summary reports work that actually reached status 'T'.

Duplicates are dropped

If dsarch reports every member as already backed up under a different tar, the placeholder record and its lists are removed.

In an -A 3 run both phases tar. A list created this run is tarred right away, and any 'N' record left over from before is tarred too.

-A 4 Deep Dive: Transferring to Quasar

For each batch of tar files

1. Walk the 'T' records, accumulating tars until the batch hits 90 GB
2. Confirm each tar still exists on gdex-glade; if not, reset it to 'N'
3. Hand the whole list to dsglobus as one transfer — Drdata first, then Backup
4. Wait for both transfers to report finished
5. Delete the local tars and set every record to status 'A'

All or nothing. If either copy fails, no local tar is deleted and no record advances — the whole batch is retried by a later run.

Why batch to 90 GB?



18 tar files × 5 GB = one 90 GB dsglobus transfer

dsarch AQ can upload a tar as it builds it, but only one at a time. dsquasar calls dsglobus with the whole list, so a batch costs one request, not eighteen.

Source and destination paths

```
from /data/gdex-quasar/d123456/<tar>      (Web)
     /decsdata/gdex-quasar/d123456/<tar>  (Saved)
to   /d123456/G001/<tar>
```

Single worker only. Transfers claim tars without dataset locks, so -A 4 never runs two workers.

Re-Backing Up Files That Changed

A file already on tape can be replaced on disk: a corrected version, a re-processed year. The tape copy is then stale. `-c` looks past the “never backed up” test and picks up files whose GDEXDB record changed recently.

```
dsquasar -t d123456 -A 1 -c 30
```

Only meaningful with `-A 1` — the gather step is what decides which files land in the input lists. Must be greater than 0.

What changes

Selection

adds files with a recent change date, even if they already have a backup record

Tar name

gets a `_changed` suffix so the re-backup is obvious on tape

Old copy

stays on Quasar — nothing is deleted; the new tar is an additional version

The four-way gather

Each dataset is walked for up to four input lists, and `-c` widens each of them:

List	Contents
Web / new	Web files with no backup record
Web / chg	Web files changed within ChangeDays
Saved / new	Saved files with no backup record
Saved / chg	Saved files changed within ChangeDays

Pick ChangeDays to match your run cadence

A large window re-tars files that were already re-tarred on the previous run, burning tape for nothing. Roughly one gather interval plus a safety margin is the useful range.

Group flags still apply. A group with backflag 'N' is skipped even for changed files.

-A 8 Integrity Check • -A 16 Statistics

8 -A 8 — does tape match the database?

Walks every bfile record with status 'A' for the selected datasets, asks the Quasar end point for the real file size, and compares it against the size recorded in GDEXDB.

```
dsquasar -a -A 8 -e
```

sizes agree record is left alone; the tape copy is trusted

sizes differ status is reset 'A' → 'N' and the file is re-listed, re-tarred and re-sent on the next -A 3 / -A 4

file missing same reset — the backup is simply redone from the archive

16 -A 16 — how much is on tape?

Reports, per dataset and as a grand total, what is already backed up and what is still waiting. No files are touched — it is a pure read of GDEXDB.

```
dsquasar -a -A 16 -E
```

Section	What it shows
Backed up	count and size already on Quasar
Ready	count and size still waiting
Tar files	bfile records by status: N / T / A
Per dataset	one line each, plus a summary row

Pair with -e or -E. Statistics are most useful mailed to the specialist.

-d PBS — Running as a Batch Job

Backing up the whole archive is far too big for a login session. -d turns the run into a delayed job: dsquasar writes a dscheck record and exits, and the dscheck daemon submits it to PBS when a slot is free.

```
dsquasar -a -A 3 -e -b -d PBS          # the usual cron line
```

1 Cron fires

dsquasar -a -A 3 -e -b -d PBS runs on the login host, in background mode.

2 Size the job

The detail file count is turned into a cpu count, and PBS options are written: walltime + select=1:ncpus=N:mem=Ngb.

3 dscheck record

A record is inserted keyed on command + specialist + argv + workdir. The argv is left unchanged, so the next identical cron line is recognised as a duplicate and blocks.

4 Daemon submits

dscheck picks the record up, submits to PBS, and dsquasar re-runs there — reading its process count back from the reserved ncpus.

5 Retry on failure

TRYLMTS['dsquasar'] = 3: a failed run is resubmitted up to three times before the record is reported as failed.

-d takes optional HostName and TryCount. -d PBS is the normal form; a bare -d uses the configured default batch host.

How Many CPUs Does the Job Ask For?

Tarring and uploading have very different shapes, so they are sized independently and the larger request wins — the phases run one after the other inside a single PBS allocation, sharing one ncpus.

Tarring (-A 2, -A 3)

One child per tar file — many small, cheap units.

```
MPTLIMIT = 200    tar files per process
MPTMAX   = 12     process ceiling
```

A small divisor with a high cap: lots of tar work should spread wide.

Uploading (-A 4)

One child per 90 GB batch — few large, network-bound units.

```
MPBLIMIT = 900    tar files per process
MPBMAX   = 4      process ceiling
```

A large divisor with a low cap: more Globus streams do not move more bytes.

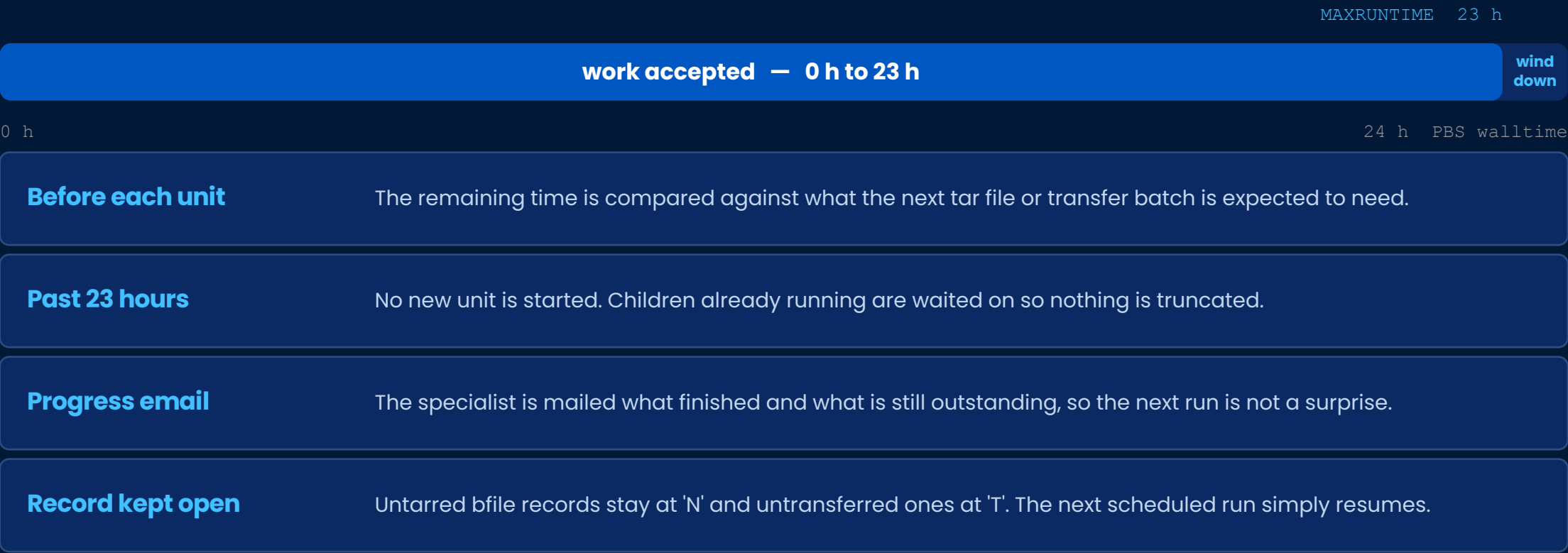
Why the two sets of numbers

Both phases once shared a single limit. Tarring came out under-provisioned while uploading reserved cpus it could never use — cpus that PBS charges for and that other jobs then cannot have. Splitting the constants fixed both ends.

-m sets the count by hand. Interactively it is whatever you pass (default 1). In delay mode the count travels through the reserved PBS ncpus instead of the command line, so -m on a -d run is ignored.

Stopping Cleanly Before PBS Kills the Job

The queue gives the job 24 hours. A tar file killed halfway through leaves a partial file on disk and a bfile record that claims work is in progress. So dsquasar watches the clock itself.



The guard only runs inside a real batch job; a foreground run is left to the operator.

-W / -w — When One Batch Job Hangs

The -A 3 gather runs from cron several times a day, so a crashed job restarts quickly — dead locks are taken over and dscheck retries. The gap was a job that is alive but making no progress: it holds the duplicate key and every later cron line simply blocks.

Stall test

A repeat submit inspects the running job. Both conditions must hold:

```
runtime > WORKGRACE (6 hours)
dcount / fcount <= MINWDONE (1 %)
```

Six hours in with under one percent of its recorded work done, the job is treated as stalled and an extra worker is submitted alongside it — up to -W MaxWorkers, default 2.

```
dsquasar -a -A 3 -b -d PBS -W 2
```

Pinned to a single worker when locking is off. Extra workers rely on dataset locks to stay off each other's files, so -l N and the transfer actions (-A 4, 6, 7) always run one worker.

Slot numbering and direction

-w N is appended to the extra worker's own command line. Since argv is the only differing key field, that gives it a dscheck record of its own instead of colliding with slot 1.

Slot	Behaviour
slot 1	plain argv — repeat cron still blocks as a duplicate
odd	walks datasets in ascending dsid order
even	walks datasets descending, from d999999

A pair of workers therefore meets in the middle rather than fighting over the same dataset lock from the same end of the list.

Not a fix for one huge dataset. Locks are per dataset, so a second worker cannot help inside the dataset that is actually stuck.

Keeping Two Runs Off the Same Files

Every mutual exclusion in dsquasar is a dataset lock — a pid and host stamped on the dataset record, claimed atomically and taken over automatically when the owning process is dead. There is no lock at the tar-file level.

Lock, then work

The dataset is locked before its files are gathered or its tars are built, and unlocked as soon as that dataset is finished.

Re-check under the lock

Work is listed before the lock is held, so the list can be hours old. Each candidate is re-read under the lock — still status 'N'? — before a tar is spent on it.

Multi-dataset tars

One tar may hold files from many datasets. All are locked before tarring and released once it is dispatched; if any is held elsewhere, the tar waits.

Tar names are safe

The predicted bid is only a guess. The insert uses AUTOID and the tar is renamed if the database returns a different id.

Operator controls

Option	Effect
<code>-l Y</code> <code>-l N</code>	Lock datasets before backing them up. Y is the default; N skips locking entirely and forces a single worker.
<code>-u</code>	Clean up backup locks left behind on the named datasets. Dataset IDs are required — not valid with <code>-a</code> .

Locks left by a killed process are reclaimed automatically; `-u` is for the cases where you want them gone now.

-e / -E — Mail to the Specialist

A batch run leaves no one watching the screen, so dsquasar mails the outcome to the dataset specialist. Two levels of detail.

-e Summary

Totals only: how many files and bytes were backed up, and how much is still waiting.

-E Detailed

The same totals plus a per-dataset breakdown — which datasets moved, which are still behind.

Typical summary mail

```
Subject: dsquasar backup statistics

Backed up : 148,392 files    612.4 GB    in 128 tar files
Ready      :  22,105 files    91.7 GB
Tar files  : N = 18    T = 6    A = 128

d123456    12,004 files    58.2 GB    backed up        2,110 files ready
d999001     1,338 files     9.9 GB    backed up          0 files ready
```

The walltime guard mails too. A batch job that stops at 23 hours sends a progress note listing what finished and what is left, so an incomplete run is visible rather than silent.

Option Reference

Selection & scope	Meaning
-a	all datasets with a backup flag set
-t ids	named datasets; SQL % wildcard allowed
-B	only work flagged 'B' (Backup only)
-D	only work flagged 'D' (Backup + Drdata)
-A bits	actions: 1 2 3 4 6 7 8 16 (default 3)
-c days	with -A 1, re-back up recently changed files
-e / -E	mail summary / detailed statistics

Either **-a** or **-t** is required — without one, dsquasar prints its usage and exits. **-B** and **-D** only narrow what a run processes; with neither, each dataset is backed up per its own backflag.

Six lines that cover most days

```
dsquasar -a -A 3 -e -b -d PBS      dsquasar -t d123456 -A 4
dsquasar -t d123456 -n             dsquasar -a -A 16 -e
dsquasar -t d123456 -A 1 -c 30     dsquasar -t d123456 -u
```

Run control	Meaning
-b	background mode, no screen output
-d [h] [n]	delay mode: submit as a PBS batch job
-m N	concurrent child processes (default 1)
-l Y N	lock datasets before backing up (default Y)
-n	show available file counts only, do nothing
-W N	max concurrent batch workers (default 2)
-w N	worker slot; set automatically, not by hand
-u	clean up backup locks on the named datasets
-h	display the help document

Where Things Live

dsquasar always runs from the Quasar backup work directory – it changes there itself, so it can be launched from anywhere.

```
$GDEXWORK/$COMMONUSER/quasar_backup
```

Work directory	Contents
<dsid>_<t>_<bid>.txt	dsarch input list, one per type per tar
<dsid>_sn..._fn....tar	the built tar, deleted after transfer
dsquasar.log	run log – first place to look
dscheck record	batch state, retries and progress counts

Globus & paths	Role
gdex-glade	source end point (archive on disk)
gdex-quasar	Backup copy on tape
gdex-quasar-drdata	disaster-recovery copy
/data/...	Web file staging path
/decsdata/...	Saved file staging path

Where the backup flags are set

dsquasar ships with rda_python_dsquasar. Backup flags are edited at gdex.ucar.edu/rda_pg_config/ (DSARCH → Dataset Information).

When Something Looks Wrong

Nothing is gathered

The dataset or group backflag is not set. Check with `dsquasar -t <dsid> -n` first.

A repeat submit does nothing

The duplicate check working: an identical command is already queued or running.

Records stuck at 'N'

Tarring never reached them — the 23-hour guard, or the dataset was locked. Resolves next run.

Records stuck at 'T'

Built but not transferred. Run `-A 4`. If the local tar is gone the record resets to 'N'.

Sizes disagree with tape

Run `-A 8` — mismatched records reset to 'N' and are redone from the archive.

Locks left on a dataset

Dead-process locks are taken over automatically. To clear now: `dsquasar -t <dsid> -u`

Statuses only ever move forward on success. Every backwards step — $T \rightarrow N$, $A \rightarrow N$ — is deliberate, and just means the work is queued to be done again.

Getting Data Back Off Tape

Backing up is only half the point. Because every tar is recorded in GDEXDB with the datasets and files it contains, recovery is a lookup rather than a search of the tape.

Restore through dsarch

dsquasar writes the backup; dsarch reads it back. The -RQ action pulls a tar from Quasar and unpacks the requested files.

```
dsarch <dsid> -RQ ...
```

The bfile record supplies the tar name, the end point and the path, so the operator does not have to know where on tape the file sits.

Two independent copies

Every tar goes to both end points by default. Drdata is written first, so an interrupted run still leaves the DR copy complete.

End point	Copy
gdex-quasar	primary Backup copy
gdex-quasar-drdata	disaster-recovery copy

No separate recovery tool

Nothing extra to install: the same dsarch that stages files for the backup also reads them back. Point -RQ at a dataset and file, and it fetches the right tar from the right end point and unpacks only what you asked for.

Verify before you need it. A scheduled -A 8 run is what turns “we have backups” into “we have backups that match the archive”.

Six Things Worth Remembering

1

dataset.backflag enrolls

Setting that one field puts a dataset under backup; dsgroup.backflag refines it. -t narrows a run, it does not override a flag.

2

-A 3 is the working default

Create the input lists and build the tars in one pass. -A 4 ships them. -A 7 does both.

3

Status is the whole story

N listed, T tarred, A archived. Backwards moves are deliberate and simply requeue the work.

4

Tar size is the economics

5 GB target tars batched into 90 GB transfers is what makes tape and Globus efficient.

5

The dataset lock is the only lock

Everything about concurrency — workers, re-checks, multi-dataset tars — is built on it.

6

Nothing is lost by stopping

The 23-hour guard, the retry limit and the resumable statuses mean an interrupted run just continues later.

Questions?

`dsquasar` — Backing Up the GDEX Archive onto Quasar

`dsquasar -h` • for the full help document

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